Six math clues · answer key

Tungsten Carbide

Age 8 / Grade 2 stretch. Offer tens-and-ones drawings, repeated addition, or oral reasoning.

Math 1 · 8

6 + 2 = 8. The pattern grows by 2 each time.

Math 2 · 20

15 + 5 = 20. The pattern grows by 5 each time.

Math 3 · 40

30 + 10 = 40. The pattern grows by 10 each time.

Math 4 · 14

12 + 2 = 14. The pattern grows by 2 each time.

Math 5 · 30

25 + 5 = 30. The pattern grows by 5 each time.

Math 6 · 70

60 + 10 = 70. The pattern grows by 10 each time.

Age 8 / Grade 2 math. Six skip-counting patterns within 100, with steps 2, 5 and 10. Say counts aloud or draw equal groups.

Reading & writing · answer key

Tungsten Carbide

Reading · 1

Hard is resistance to scratches and dents. Tough is resistance to cracking and breaking. One does not guarantee the other.

Reading · 2

Chalk snapped, while the rubber eraser bent back. This compares how they respond to bending, not their scratch hardness.

Reading · 3

A hard carbide tip resists wear. A tougher steel body supports it and takes bumps.

Writing

Accept one sensible command, spoken, dictated or written. Example: Ask Big to hold the soap and gently drag the craft stick across it.

Age 8 / Grade 3 reading and Grade 2 math. A grown-up leads every step of the optional household test; the child watches, predicts and records. Work over a tray and keep faces back. Use only soap, a smooth wooden craft stick, chalk and a flexible rubber eraser. The grown-up holds the soap and gently drags the craft stick across it, then separately bends chalk gently over the tray and bends the eraser. Stop if extra force is needed; keep crumbs on the tray. No knives, nails, glass, ceramic tiles, files, drills, saws or sharp tools; use nothing that makes shards. Never touch real saw blades or drill bits. Do not scratch with a pen tip or any tungsten carbide. WC is tungsten joined with carbon; carbide is manufactured, rather than mined as a finished tool material. Pressing and sintering usually use a cobalt binder, not household glue. Hardness and toughness vary with grain size, binder and grade. The bench gives qualitative comparisons, not measured tests or a promise that every bump chips carbide. Its scratch contact is a carbide point; on carbide it leaves no visible mark under the shown light contact. Its bump is strong enough to reveal different responses. The chalk/eraser activity is a separate bending comparison, not a ranking of toughness by measurement. Materials differ in softness as well as brittleness. Tools stay resting in pictures. Welcome dictation and a pause at any time.

Accept a spoken command, dictation, invented spelling or writing. Big is the grown-up in the story. The grown-up handles all test items; the child predicts, watches and records.

Four workshop clues

Tungsten Carbide

Quiz 1. What does hard mean?

Hard means resisting scratches and dents. A very hard material can still chip or crack.

Quiz 2. What does tough mean?

Tough means resisting cracks or breaks from bumps, bends or drops. Tough and hard are different properties.

Quiz 3. Why put a carbide tip on a steel body?

The small carbide tip resists wear. The tougher steel body supports it and takes bumps without cracking as easily.

Quiz 4. Where does tungsten carbide hide in many ballpoint pens?

Many ballpoint pens use a tiny tungsten-carbide ball at the tip. The hard ball rolls ink onto paper and resists wear.

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